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# Elements, Compounds & Mixtures Worksheet

**Part 1:** Read the following information on elements, compounds and mixtures. Fill in the blanks where necessary.

**Elements:**

- A pure substance containing only one kind of _____.
- An element is always uniform all the way through (homogeneous).
- An element _____ be separated into simpler materials (except during nuclear reactions).
- Over 100 existing elements are listed and classified on the _____.

**Compounds:**

- A pure substance containing two or more kinds of _____.
- The atoms are _____ combined in some way. Often times (but not always) they come together to form groups of atoms called molecules.
- A compound is always homogeneous (uniform).
- Compounds _____ be separated by physical means. Separating a compound requires a chemical reaction.
- The properties of a compound are usually different than the properties of the elements it contains.

**Mixtures:**

- Two or more _____ or _____ NOT chemically combined.
- No reaction between substances.
- Mixtures can be uniform (called _____) and are known as solutions.
- Mixtures can also be non-uniform (called _____).
- Mixtures can be separated into their components by chemical or physical means.
- The properties of a mixture are similar to the properties of its components.

**Part 2:** Classify each of the following as elements (E), compounds (C) or Mixtures (M). Write the letter X if it is none of these.

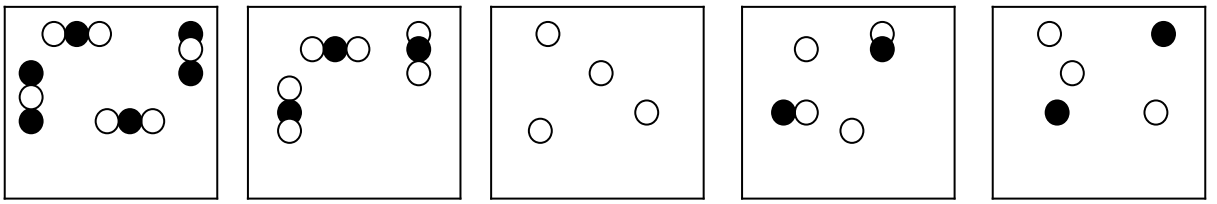
___Diamond (C) ___Sugar ($C_6H_{12}O_6$) ___Milk ___Iron (Fe)
___Air ___Sulfuric Acid ($H_2SO_4$) ___Gasoline ___Electricity
___Krypton (K) ___Bismuth (Bi) ___Uranium (U) ___Popcorn
___Water ($H_2O$) ___Alcohol ($CH_3OH$) ___Pail of Garbage ___A dog
___Ammonia ($NH_3$) ___Salt (NaCl) ___Energy ___Gold (Au)

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___Wood ___Bronze ___Ink ___Pizza
___Dry Ice ($CO_2$) ___Baking Soda ($NaHCO_3$) ___Titanium (Ti)
___Concrete

**Part 3:** Match each diagram with its correct description. Diagrams will be used once.



**A** **B** **C** **D** **E**

___1. Pure Element – only one type of atom present.

___2. Mixture of two elements – two types of uncombined atoms present.

___3. Pure compound – only one type of compound present.

___4. Mixture of two compounds – two types of compounds present.

___5. Mixture of a compound and an element.

**Part 4:** Column A lists a substance. In Column B, list whether the substance is an element (E), a compound (C), a Heterogeneous Mixture (HM), or a Solution (S). (Remember a solution is a homogeneous mixture.) In Column C, list TWO physical properties of the substance.

| Column A | Column B | Column C |
|---|---|---|
| 1. Summer Sausage | | |
| 2. Steam | | |
| 3. Salt Water | | |
| 4. Pencil lead (Pb) | | |
| 5. Dirt | | |
| 6. Pepsi | | |
| 7. Silver (Ag) | | |
| 8. Toothpaste ($Na_2HPO_4$) | | |
| 9. A burrito | | |
| 10. Italian Dressing | | |
| 11. Chicken Soup | | |

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| | | |
|---|---|---|
| 12. Lemonade | | |

## Elements, Compounds & Mixtures Worksheet

**Part 1:** Read the following information on elements, compounds and mixtures. Fill in the blanks where necessary.

**Elements:**

- A pure substance containing only one kind of __**atom**____.
- An element is always uniform all the way through (homogeneous).
- An element __**cannot**___ be separated into simpler materials (except during nuclear reactions).
- Over 100 existing elements are listed and classified on the _**Periodic Table**_.

**Compounds:**

- A pure substance containing two or more kinds of __**atoms**__.
- The atoms are ___**chemically**___ combined in some way. Often times (but not always) they come together to form groups of atoms called molecules.
- A compound is always homogeneous (uniform).
- Compounds ___**cannot**___ be separated by physical means. Separating a compound requires a chemical reaction.
- The properties of a compound are usually different than the properties of the elements it contains.

**Mixtures:**

- Two or more __**elements**___ or ____**compounds**__ NOT chemically combined.
- No reaction between substances.
- Mixtures can be uniform (called __**homogeneous**___) and are known as solutions.
- Mixtures can also be non-uniform (called ____**heterogeneous**____).
- Mixtures can be separated into their components by chemical or physical means.
- The properties of a mixture are similar to the properties of its components.

**Part 2:** Classify each of the following as elements (E), compounds (C) or Mixtures (M). Write the letter X if it is none of these.

_**E**_Diamond (C) _**C**_Sugar ($C_6H_{12}O_6$) _**M**_Milk

_**E**_Iron (Fe)

_**M**_Air _**C**_Sulfuric Acid ($H_2SO_4$) _**M**_Gasoline

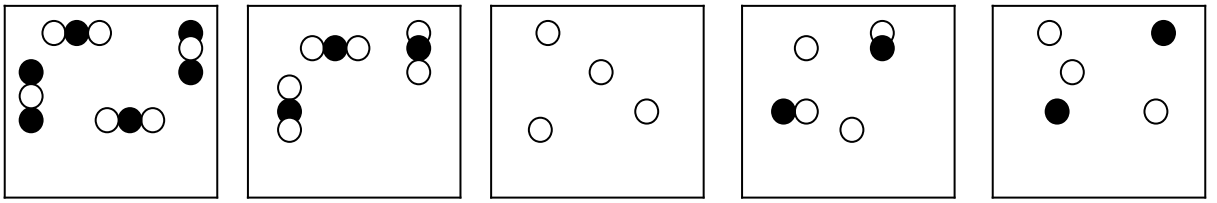
_**X**_Electricity

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_**E**_Krypton (K) _**E**_Bismuth (Bi) _**E**_Uranium (U) _**M**_Popcorn
_**C**_Water ($H_2O$) _**C**_Alcohol ($CH_3OH$) _**M**_Pail of Garbage _**M**_A dog
_**C**_Ammonia ($NH_3$) _**C**_Salt (NaCl) _**X**_Energy _**E**_Gold (Au)
_**M**_Wood _**M**_Bronze _**M**_Ink _**M**_Pizza
_**C**_Dry Ice ($CO_2$) _**C**_Baking Soda ($NaHCO_3$) _**E**_Titanium (Ti)
_**M**_Concrete

**Part 3:** Match each diagram with its correct description. Diagrams will be used once.



**A** **B** **C** **D** **E**

_**C**_1. Pure Element – only one type of atom present.

_**E**_2. Mixture of two elements – two types of uncombined atoms present.

_**B**_3. Pure compound – only one type of compound present.

_**A**_4. Mixture of two compounds – two types of compounds present.

_**D**_5. Mixture of a compound and an element.

**Part 4:** Column A lists a substance. In Column B, list whether the substance is an element (E), a compound (C), a Heterogeneous Mixture (HM), or a Solution (S). (Remember a solution is a homogeneous mixture.) In Column C, list TWO physical properties of the substance.

| Column A | Column B | Column C |
|---|---|---|
| 1. Summer Sausage | **HM** | **Chunky, Brown** |
| 2. Steam | **C** | **Gas, Hot** |
| 3. Salt Water | **S** | **Liquid, Clear** |
| 4. Pencil lead (Pb) | **E** | **Grey, Solid** |
| 5. Dirt | **HM** | **Brown, Solid** |
| 6. Pepsi | **HM** | **Brown, Liquid** |
| 7. Silver (Ag) | **E** | **Silver, Solid** |
| 8. Toothpaste ($Na_2HPO_4$) | **C** | **White, Thick** |
| 9. A burrito | **HM** | **Multi-colored, Solid** |
| 10. Italian Dressing | **HM** | **Liquid, Greasy** |

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| | | |
|---|---|---|
| 11. Chicken Soup | **HM** | **Liquid/Solid, Brown** |
| 12. Lemonade | **S** | **Yellow, Liquid** |